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Opportunities and challenges of toolpath-based 3D printed textiles: insights from Digital Design and Craft course

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Advances in additive manufacturing (AM), particularly Fused Deposition Modeling (FDM), have introduced new possibilities for textile and fashion design through toolpath-based fabrication. This paper examines the opportunities and challenges of 3D-printed textiles, drawing on insights from five iterations of a Digital Design and Craft course, in which 101 students designed and produced wearable artifacts using computational workflows. This paper examines students' projects, analyzes application types, discusses assembly techniques and post-processing strategies, and highlights how traditional textile manufacturing inspires the toolpath structures. The paper further discusses the opportunities of 3D printed toolpath-based textiles, such as supporting sustainability and enhancing personalization through accessible, low-cost technologies. Challenges include differences in tactility relative to traditional textiles, printer build-size constraints that require assembly, and reliance on post-processing to achieve three-dimensional forms. By framing these findings within an educational context, this paper highlights both the creative potential and current barriers of 3D printed textiles and suggests directions for future research, tool development, and design pedagogy.

Keywords: personal fabrication; computational fabrication; toolpath manipulation; computer science education

1. Introduction

Advances in Additive Manufacturing (AM) technologies have enabled new possibilities and opportunities for fashion and textile designers. These technologies can be used to enhance performance, improve fit and customization, and generate new aesthetics for garments and wearables (Dilek et al., 2021; Keefe et al., 2022; Sitotaw et al., 2020). Moreover, they offer a pathway toward more sustainable manufacturing processes, enabling the production of zero-waste products, improving recyclability, and promoting on-demand manufacturing (Khajavi, 2021). Among AM's

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various technologies, Fused Deposition Modeling (FDM) stands out for its accessibility, low cost, and minimal post-processing required for printed objects (Elsonbaty et al., 2024).

Traditionally, 3D printing involves creating a 3D model in a Computer-Aided Design (CAD) environment or retrieving one from an online library. This model is then processed by the Slicer software to generate the toolpath and a G-code file containing instructions for the printer. This workflow is relatively simple and straightforward and can generate volumetric 3D models, but it offers very limited control over the toolpath and its parameters, as the slicer algorithm automatically determines them. This limitation is particularly critical for 3D printing textile-like geometries, since controlling and manipulating the toolpath and printing parameters can produce a wide range of shapes, functional structures, mechanisms, and aesthetic outcomes that are not possible with the standard workflow.

Indeed, toolpath manipulation on FDM printers is adopted by designers and researchers to create a variety of textile-related projects. This includes generating a weaving pattern from 3D-printed strands (Partsch et al., 2015, p. 3; Takahashi & Kim, 2019), printing quasi-static, tulle-like structures leveraging under-extrusion (Forman et al., 2020), and 3D printing spacer fabrics with a construction similar to knitted spacer fabrics (Halbrecht et al., 2023). These examples demonstrate that the printed strands can be manipulated in space as they are deposited from the printer's nozzle to build a structure, resembling how yarns are manipulated in traditional textile manufacturing. In these examples, the fabric is built upright on the build platform, enabling the production of 3D geometry. However, this approach is limited by gravity, which affects the geometry and the possible design of textile structures since support is required for overhang geometries. Alternatively, the textile can be printed flat on the build platform, creating a thin 2D sheet of material, eliminating the need for support. This approach enables countless design opportunities for different structures, as the printed traces are supported by the platform (Chatterjee & Ghosh, 2020). By printing a few layers of these continuous overlapping paths, it is possible to produce thin, 2D net-like material sheets. These nets, especially when printed with flexible filaments such as Thermoplastic Polyurethane (TPU), exhibit textile-like qualities such as drape, flexibility, and light weight, which are comparable to traditional textile materials (Zhan et al., 2026). These 3D-printed net-like materials are being used by hobbyists, makers, and global brands such as Nike, which uses this technology to produce shoe uppers and performance sports bras (*Nike FlyWeb Is the Future of Performance Apparel — NIKE, Inc.*, n.d.).

In the context of academic graduate curriculum in design and fabrication, the question of how designers can use advanced technologies and explore new aesthetic and performance applications is fundamental (Buechley et al., 2022; Montero, 2018; Oktan & Vural, 2022). To creatively explore the design possibilities of textile 3D printing, it is necessary to use computation to manipulate the toolpath. This manipulation is done by generating a set of rules, defining parameters, and creating workflows that output the printer's toolpath. It is also important to understand the printer's constraints and physical limitations, as this printing technique is exploratory and often requires working in close proximity to the printer. However, most designers are not trained in coding and algorithmic thinking, and they lack the fundamental knowledge of software development needed to creatively leverage new AM technologies beyond standard workflows. To address this gap, the Digital Design and Craft (DDaC) course was developed in the Faculty of Architecture and Town Planning at the Technion-Israel Institute of Technology. The objective of this course is to teach designers basic coding principles and to demonstrate how they can be applied for fabrication. The course focuses on

building software applications that generate toolpaths and produce G-code files that can be loaded into a 3D printer and fabricated.

The DDaC course is part of the Industrial Design graduate program. Most students taking this course are graduate students with some design background (e.g., architecture, industrial design, fashion design). Many of them start the course without any background in coding or computational design. The final assignment for the course is to build a wearable product using 3D-printed parts generated via toolpath manipulation. This assignment is chosen because it gives students the opportunity to design a full product and address issues such as fit, customization, personal preferences, and style.

Over five iterations of the course, which started in its current form in 2021, 101 students have participated and successfully graduated after completing the final assignment. This paper examines the course's final project assignments, analyzes the types of products students chose to create and the materials used, discusses assembly techniques and post-processing strategies, and highlights trends and innovative approaches. The paper further explores the cultural and traditional textile inspirations and motifs embedded in the students' projects, adding an extra layer of cultural significance.

Examining the final assignment projects highlights the potential of personal fabrication and the future of textile and fashion design, including opportunities and challenges that may inspire future research and development of new technological and design tools. Aside from an example of a didactic approach for training designers in academic settings to learn new fabrication technologies, this paper contributes to the positioning of 3D printed textiles as a creative medium for personal fabrication. By demonstrating the richness of ideas, applications and approaches in students' projects, this paper reveals the creative potential of toolpath manipulation for wearable artefacts.

2. The DDaC Course

The DDaC is an annual elective course in the graduate program in industrial design. In its current form, the course has had five iterations, with an average of 20 students per year. Although it is a graduate-level course, undergraduate students are also invited to enroll.

Students enrolled in this class usually have a design or engineering background. Coding experience and knowledge of computational design tools, such as Grasshopper for Rhino, vary widely; most students have no coding experience, and some, mainly students from architecture backgrounds, are experienced. The course assignments are designed to accommodate different levels by allowing each student to express themselves to the level that they possess. This results in varying quality levels across projects, as beginners often struggle more with coding.

The course is structured around three streams. The first involves paper presentations that each student gives once during the semester. At the beginning of the course, students pick a paper from a provided list to present to the class during one of the course meetings. The papers on this list are recent computational fabrication studies, usually presented at top HCI conferences such as Computer-Human Interaction (CHI), the Symposium on User Interface Software and Technology (UIST), and the Symposium on Computational Fabrication (SCF). The purpose of this presentation is to familiarize students with this research theme and showcase the creative potential of computational fabrication. The second stream consists of lectures given by the course instructor on fabrication, FDM printing,

and examples of creative toolpath manipulation. There are also technical lectures on the printer's construction, G-code language, and 3D printing file formats. The third stream includes live tutorials and code demonstrations, typically conducted by the course teaching assistant. These tutorials are walkthrough examples of code that serve as the fundamental building blocks for the course assignments. By providing these examples, students without coding experience can creatively modify parameters, hack the code by adding their own modifications, and tweak the functionality.

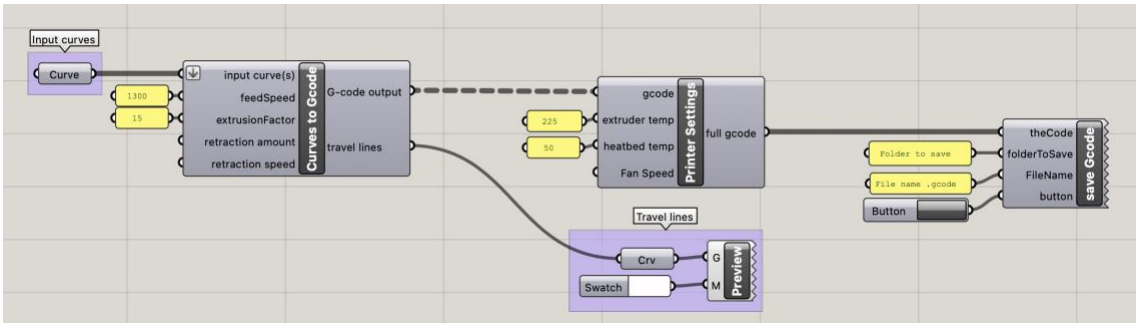


Figure 1. The Grasshopper components for translating curves into executable G-code.

Since DDaC is an introductory course for designers learning coding, it is important to provide a holistic overview of algorithmic thinking rather than focusing on a single tool. Therefore, the course is divided into two complementary parts. In the first part, students work with scripting languages such as Python or Processing. This section explains basic coding principles, such as variables, loops, and if statements. It is also dedicated to learning G-code, including its structure, syntax, and commands. The second part of the course centers on visual coding through designing textile structures in Grasshopper for Rhino. The Grasshopper environment offers greater flexibility and is generally easier for designers, as it allows them to use the Rhino interface for direct geometry manipulation (e.g., drawing curves in Rhino) with parametric design. Using the example given in class, students design various toolpath-based textile structures, considering factors such as toolpath continuity, adhesion to the build plate and between layers, the mechanical properties of the resulting structures, aesthetics, and the effects of various parameters on the printers' behavior and extruded material. The translation from curves to G-code is done by a series of Grasshopper components provided to students. These components allow the input of any set of curves and output a ready-to-use G-code file. The components allow controlling printing parameters and previewing the travel lines (Figure 1). Eventually, insights from the first part of the course can be applied by incorporating Python code components into the Grasshopper definition to streamline workflows and expand the design space.

3. Final Assignment Project Overview

The course final assignment is called "G-code to Wear." The task is to create a wearable product using 3D-printed parts generated through code-driven toolpath manipulation. The wearable item can be an accessory, a garment, or any other object that contacts the body. Students are encouraged to use 3D-printed parts made with the standard slicing workflow or to combine other materials that are not 3D-printed, as long as most of the work is done with code. They could also use any method for bonding, assembling, and post-processing. To make the project easier, students could build it on a smaller scale. Students had about three weeks to complete this assignment. Table 3 in the appendix summarizes all 101 projects completed in the last 5 years for the DDaC course. The table provides the students' names

(in initials), project names, and descriptions. It also includes an analysis of each project: the product category, assembly method, and materials used.

Below is a detailed analysis of the projects:

3.1. Scale

Although the student had the option to produce a reduced-scale version, which required less effort, only one project (project number 40 in Table 3) was produced on a smaller scale. This reflects students' enthusiasm for building something they can wear and use, even if it involves extra work. It is worth noting that although only one project was done on a reduced scale, in many other projects, the students chose to produce small-sized objects (for example, wallets and jewelry).

3.2. Diversity

Analyzing the project and the applications used to create the wearable reveals the variety of ideas, applications, and interpretations among the students. Their projects were divided into nine categories based on their theme or use. Within each category, there is a great variety of applications. Table 1 summarizes the project categories and the type of product in each.

Table 1. Summary of application types.

Category name	Product types
Accessories	bracelets (5), collars (4), necklaces (3), gloves (3), belts (2), watch straps (2), tie, sleeve, ring, earrings, boutonniere, sunglasses strap, niqab, hand fan, shawl
Hats	fascinators (4), wide brimmed hats (3), yamakas/prayer hats (3), cap hats (2), kerchiefs (2), bucket hats (3), hat-nets (2) costume hat, visor, wig
Bags	wallets (4), tote bags (5), market bags (2), Clutch bag, sling bag, pouch, beach bag
Tops	vests (5), bralettes (2), bra, shawl, corset (1), apron
Footwear	sandals (4), sneakers (3), slider, sock sleave
Face wear	glasses frames (4), Covid mask
Design for Pets	dog harnesses (2), dog collar
Skirts	skirts (2)
Misc.	Medical hand splint, a method for patching holes, reinforcing riding pants by printing, printing of wood paste, auxetic knee protector.

3.3. Textile structures and patterns

Using computational code to generate the toolpath offers many options for creating complex patterns and curve arrangements that resemble textile materials with various functions and aesthetics. The basic structure includes an orthogonal grid, where one layer produces longitudinal lines and a subsequent overlay creates horizontal lines. The printed material fuses at the intersections of these layers, giving the fabric strength and durability. Building on these principles, different variations of this construction are used. For example, Gali Cnaani created a wearable collar based on a curvilinear grid. The collar was printed as six identical pieces with the grid deformed using a sine-wave algorithm. An iron and backing paper were used to fuse the pieces together (Figure 2). Some students used references from traditional textile production to generate printed structures: Dana Shay created a long-brimmed hat from PLA. She was inspired by patterns from woven straw hats and incorporated them into the printed pattern (Figure 3). Luba Texler created a vest with an overlapping circular pattern resembling a chainmail structure. This pattern allows for visual continuity, as the different

printed pieces overlap, hiding the seams. The pieces are connected to each other by adding material with a 3D printing pen (Figure 4).

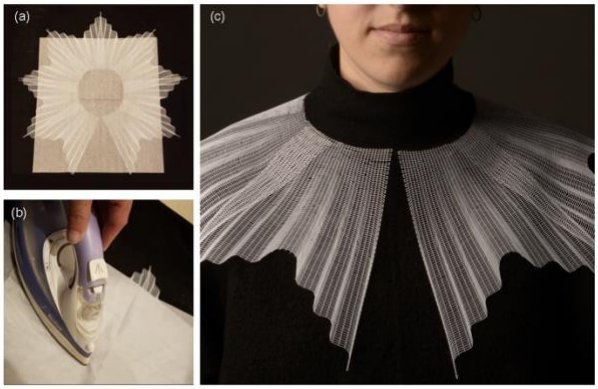


Figure 2. A collar by Gali Cnaani, printed using a curvilinear grid. The collar is made of six pieces (a), connected by fusion using an iron (b). The flexible collar drapes on the body (c).

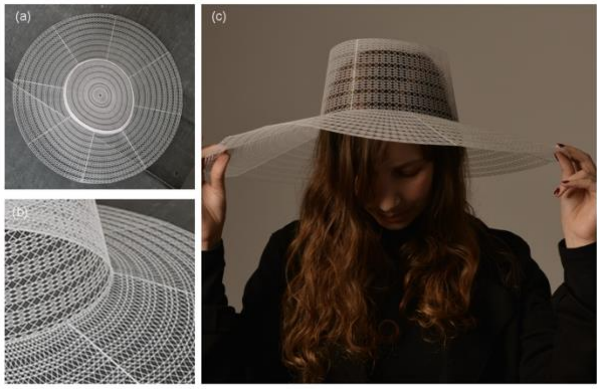


Figure 3. A long-brimmed hat by Dana Shay made from PLA. The hat is made from 11 pieces (a), glued together using Superglue. The textile structure is inspired by straw hat weaving (b,c).

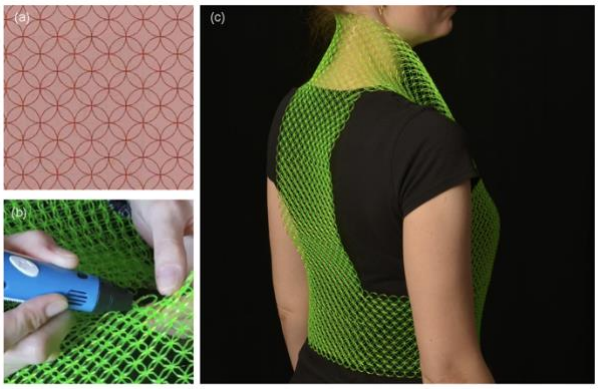


Figure 4. A chainmail vest created by Luba Texler using a pattern of overlapping circles (a). the printed pieces are connected using a 3D pen (b). The chainmail pattern helps in eliminating the seams and creating an appearance of a continuous fabric.



Figure 5. A tote bag by Anastasia Zilbert is composed of interwoven pieces. Flower-like printed piece has 4 flaps (a), that are inserted into notches in a circular piece (b). the assembled pieces generate a lace-like structure for the bag (c).

Another common theme is lace. Since the printed paths are thin and can generate complex morphologies and interwoven patterns, some students were inspired by lace to create printed tops and bags. Anastasia Zilbert created a tote bag composed of lace-like interwoven pieces. Using two basic shapes, one with four flaps and the other with notches, the 3D shape of the bag can be formed. The handles have a different lace pattern. Together, the pieces form a textured bag with delicate lace perforations (Figure 5). Another lace-inspired project is the Computational Bralette by Maya Kaplan. The bralette combines a variety of printed structures positioned according to their location and function on the body. For example, some areas are stretchable, and others are stiff. She used overprinting to connect the pieces together, using the 3D printer as a sewing machine. This is done by printing zigzag-stitch lines over two pieces to join them (Figure 6).

Inbar Petel-Niv used fringes in her skirt design. The idea of using fringes was discovered by accident, as the printer jammed mid-print while she was printing a chevron grid. The first layer, composed of parallel lines, was joined by the second layer only partially, leaving the strands to move freely as fringes. The student then decided to use the fringes in her final skirt design (Figure 7).



Figure 6. 3D printed bralette by Maya Kaplan. The pattern pieces (a) are connected using overprinting of a stitch pattern (b). The bralette is designed with different structures and densities, mapped to requirement of different zones (c).



Figure 7. A fringe skirt designed by Inbar Petel-Niv. The printed pattern is manually sewn onto the skirt fabric (a). Fringes are created by printing a layer of horizontal lines and a partial layer of crossing lines (b). The final skirt uses those fringes as decoration (c).

3.4. Materials

The most common materials used for fabrication are Polylactic acid (PLA) and TPU. TPU is a flexible material that, when printed as a thin toolpath-based textile, provides flexibility, elasticity, and drape. However, TPU is more challenging to print because it tends to jam in the extruder at high nozzle pressure. PLA is easier to print, but it is a rigid material with limited flexibility when printed as thin paths. 85% of projects used TPU, and 22% used PLA. In one case, both materials were used.

3.5. Assembly methods

One of the challenges for producing full-sized wearable items is their size, which is usually larger than the printer bed area. This limitation sparked different creative solutions for attaching, assembling, and connecting printed pieces to form the final projects. These methods were grouped by general themes, but within each group, there is considerable diversity of techniques and approaches (Table 2). Some students relied on mechanical connections of parts. This was done by using fasteners such as metal rings or pins. Another method is to generate flat pieces with notches in the printed parts, allowing interweaving of parts to produce large sheets and three-dimensional shapes without seams (see Figure 5). Thermal bonding is another method for connecting pieces. This is done by melting the thermoplastic material, allowing pieces to join without adding an extra material. Students use this technique with a steam iron to fuse pieces together using a piece of baking paper to avoid direct contact (Figure 2). Another method for thermal fusing is using a soldering iron or the hot printer nozzle, pressing the tool against the material to achieve adhesion. An additional method for joining is overprinting, which involves adding material to connect pieces. This was done with a handheld 3D pen or by using the printer to print an additional path over already-printed pieces to connect them (see Figures 4 and 6). This method is similar to sewing, where the connecting thread is the printed material; however, it doesn't introduce a new material, thereby enabling recycling. Other methods include using glues such as hot-melt glue or Superglue, or sewing pieces together with a sewing machine or manually.

Table 2. Methods of assembly and connection of printed pieces. In parenthesis. The number of projects that used this method.

Connection type	Methods
Mechanical connectors	Interconnected pieces using flaps and notches (11), metal rings (5), metal pins (2)
Heat Fusion (welding)	fabric iron with backing paper (1), soldering iron (3), heated metal (1), using the printer's nozzle for welding (1)
Sewing	Winding thread, metal wire (11), machine sewing (6), hand sewing (4)
Overlay material deposition	Using the printer for connecting parts (4),3D pen (4), printing on paper (1), printing on fabric (3)
Glue	Hot-melt glue (3), super glue (2)

3.6. Morphing and forming

Since the printed textile is printed flat, it needs to be deformed to create 3D or volumetric shapes. When TPU is used, the material is flexible and can easily be deformed into a 3D surface. Some projects used origami-style folding, leveraging the material's flexibility by incorporating creases into the design. Unlike TPU, PLA is rigid and requires post-processing to form a 3D shape. Some students used heat to soften the material, then deform it into a 3D shape, either manually or by pressing it against a 3D form. Heat was applied in various ways: using a hot-air blow gun, dunking the printed pieces in hot water, or using a hair iron to heat them and then deform them.

4. Discussion

This study highlights both the emerging potential and the current limitations of toolpath-based 3D-printed textiles, as observed across five iterations of the DDaC course and the diverse body of student work produced. Notably, for many of the students this course was the first encounter with coding, and for some the first time working with the 3D printer. This demonstrate the effectiveness of the course's approach of learning by making, in which learning is achieved though iterative experimentation and generation of physical artifacts.

Altogether, the projects point to a rapidly expanding design space enabled by computational toolpath-based 3D printing , while also revealing material, technical, and practical constraints that shape its present applicability in textile and fashion contexts.

4.1. Opportunities

The breadth of students' approaches and ideas for wearable applications is evident when inspecting the full corpus of project. This is reflected in the 9 project application categories and the variety that exists within each. This variety highlights the opportunities in the expanded design space afforded by computational design and toolpath manipulation. The ability to directly control toolpaths enables the creation of textile-like structures that go beyond the constraints of traditional weaving or knitting, for example, generating shaped perforations in different zones of the textiles (as can be seen in Figures 3 and 6). Designers can generate complex geometries, gradients, and responsive structures by adjusting parameters such as path spacing, layering, and extrusion behavior. This computational approach supports a high degree of creativity, expressiveness, and experimentation, allowing designers to embed cultural references inspired by traditional textile patterns such as lace and straw weaving, or invent entirely new material aesthetics. The emergence of unexpected outcomes—such as exploiting

printing “failures” as design features—further underscores the exploratory nature of this medium. This is exemplified in the fringe skirt, where the fringes were added as a result of a failed print.

The projects also demonstrate the promise of personal fabrication. FDM printing is relatively accessible due to its low cost and widespread availability, making it feasible for individual designers, makers, and small studios. This accessibility enables customization and personalization, as garments can be tailored to individual preferences, body measurements, and stylistic choices. The accessibility and low cost of 3D printers contributed to the observed strong tendency of students to produce full-scale wearable objects, despite the additional effort required. It reflects a desire to create meaningful, user-specific artifacts rather than abstract prototypes. This suggests that 3D-printed textiles may help shift fashion production toward more individualized and locally distributed models.

4.2. Challenges

Despite these opportunities, several challenges remain. A primary limitation concerns material properties, particularly tactility. While flexible filaments such as TPU can approximate some textile-like qualities (e.g., drape and flexibility), they still do not fully replicate the softness and sensory richness of woven or knitted fabrics. Rigid materials like PLA further emphasize this gap, often requiring post-processing such as heating and forming to achieve wearable forms. This limitation is evident in the number of of applications that are not in direct contact with the skin such as hats, bags, earrings, or non-textile wearables such as jewelry and eyewear. Together, these applications comprise more than 75% of all projects.

From a fabrication standpoint, printer size constraints present a practical challenge. The limited build volume of desktop FDM printers necessitates segmenting garments into smaller components that must then be assembled. While this constraint has led to creative assembly strategies—such as interlocking systems, thermal fusion, and overprinting—it also introduces additional labor and potential weak points in the final product. These processes can complicate scalability and consistency in production.

Finally, there is a conceptual and technical limitation: many printed textiles are made as flat sheets, which do not inherently utilize the full three-dimensional capabilities of additive manufacturing. While printing fabrics as flat sheets eliminates the need for support, this method requires post-processes steps such as folding, bending, or heat-forming to create volumetric shapes. While these post-processing techniques can work well, they add variability and decrease workflow efficiency. This points to an area for future research: developing methods that embed 3D form creation directly into the printing process instead of relying on secondary deformation.

5. Conclusion

This paper has explored the opportunities and challenges of toolpath-based 3D printed textiles through the lens of a graduate computational design and digital fabrication course. The wide variety of student projects demonstrates the expressive potential of this approach, spanning diverse product typologies, innovative material uses, assembly methods, and aesthetic strategies. These outcomes highlight how computational control over toolpaths can expand the design space of textiles and enable new forms of creativity in fashion and wearable design.

Importantly, the course shows that the computational skills required for such exploration—often perceived as a barrier for designers—can be effectively taught, even to students with no prior coding experience. Through a structured combination of scripting, visual programming, and hands-on fabrication, students developed the ability to design and fabricate complex textile structures. The resulting projects reflect not only technical proficiency but also strong individual expression, indicating that coding can function as a creative medium rather than merely a technical tool.

In conclusion, while challenges remain, the convergence of computational design, additive manufacturing, and textile thinking presents a compelling direction for the future of fashion and wearable design. The DDaC course serves as a valuable model for equipping designers to engage with these technologies and bridge the gap between technical innovation and creative practice.

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Appendix A – Table 3

#	Year	initials	project name	Category	Description	Assembly method	Material	Student Type
1	2021	I.K	**La boutonnieres**	Accessories	A flower-like ornament for men's suits. The star-shaped patterns are printed and transformed into a flower shape, then attached to a printed stem to create this ornament.	thread	TPU	MSc
2	2021	Y.A	**Sroch**	Accessories	Sunglasses strap with a wavy pattern that enables flexibility and extension when needed.	no assembly	TPU	MSc
3	2021	L.A	**Flip-flops**	Footwear	Toddler's sandal with a Voronoi outsole, and a strap based on a grid pattern.	pins	TPU	MSc
4	2021	M.B	**No Covid mASK**	Face wear	A COVID mask printed in two halves with a seam in the middle, featuring a grid pattern made with Fluorescent filament.	welding with soldering iron	TPU	MSc
5	2021	S.B	**Hatnet**	Hats	A Victorian hat with a printed, lace-inspired structure. The hat is made of 5 pieces but features a radial, wavy printed structure. The see-through hat covers the head and eyes, with straps tied under the chin.	hand sewing	TPU	MSc
6	2021	O.B	**Conductive Shoe**	Footwear	A toddler's sandals feature a flat sole with a hole in the heel and a flexible, printed upper. The sandle concept is reconnecting the body to the Earth's surface.	glue	TPU	MSc
7	2021	D.B.G	**A G-code Fascinator**	Hats	A fascinator based on a circular mesh pattern and leaf-like printed pieces. The leaf-like pieces are wedged into the rounded base shape to create a flower-like appearance.	Welding with hot metal and backing paper.	TPU	PhD
8	2021	N.B	**Compact wallet design**	Bags	A wallet for a single credit card. The bottom and side walls of the wallet are printed first, then the card is inserted into the printer and the printing resumes on top of it.	no assembly	TPU	MSc
9	2021	G.S	**Magnetic eyewear**	Face wear	Eyewear frame with a pattern based on magnetic lines affected by attractors located around the eye location.	Unspecified	TPU	MSc
10	2021	D.G.G	**Ballet- hand splint**	Misc	A medical splint designed for hand support. The splint is printed flat and shaped by dipping into hot water. Its design features crossing perpendicular lines, offering both strength and breathability.	no assembly	TPU	MSc
11	2021	G.C.S	**Lace Collar**	Accessories	A Victorian lace collar inspired by the Elizabethan lace collar. The flat collar creates an illusion of a 3D structure.	ironing with a backing paper	TPU	MSc
12	2021	A.G	**Experimental Textile Surface**	Accessories	A parametric bracelet inspired by The Glitch dress by Iris Van Herpen. As the bracelet compresses on the hand, free-standing strands bend and	no assembly	TPU	MSc

					generate a flower-like appearance.			
13	2021	G.D	**3D printed head piece**	Hats	A fascinator based on a flat, rounded panel with a radial grid, decorated with a feather and tied to the head with a strap.	glue	TPU	MSc
14	2021	E.H	**Make your own yamaka**	Hats	A Yamaka with a star-of-David grid pattern. The design is based on two pieces, each with a single notch. The pieces are stitched together by hand sewing.	Hand sewing	TPU	PhD
15	2021	M.H	**PRINTED TIE**	Accessories	A printed rigid tie embedded with a collar. A hexagonal pattern or a deformed rectangular grid pattern is used. The collar is deformed from flat to a curled form using a hair iron device.	no assembly	TPU	MSc
16	2021	A.H	**Gcode to wear- shoe**	Footwear	3D-printed shoes with both the upper and sole 3D-printed. The upper is based on a transitional upper making, using a last to extract the pattern. The sole is based on a chevron pattern	sewing machine	TPU	MSc
17	2021	H.T. W	**Gee Bag**	Bags	A small tote bag based on a hexagon shape and a large ring as handles. A cotton strip is attached to the bottom to increase the bag volume two metal rings are attached to the nets using metal wires	sewing machine	TPU	MSc
18	2021	A.V	**Fully printed cap**	Hats	A baseball-style cap consists of 6 panels, a two-part strap closure, and a visor. All parts are 3D printed. Some panels are filled with an overlapping circle grid pattern, while others feature a shear-based “expanding’ pattern.	welding using the 3D printer	TPU	PhD
19	2021	M.K	**Elizabeth**	Hats	A fascinator for a flower-like appearance. The fascinator is generated using PLA and deformed into an organic shape using a hot-air blower and round objects as a preform. The printed structure is based on a grid with variable densities.	no assembly	TPU	MSc
20	2021	R.T	**Barefoot shoe**	Footwear	A sandal with a lace-like 3D printed upper. The sole is made with a rigid material, and the upper covers the whole foot up to the ankle. The fastening is done with a string at the back of the foot.	3D printing pen	TPU	MSc
21	2021	R.I	**Untitled**	Face wear	A glasses frame inspired by the drawing of Mondrian. The design is based on a silhouette filled with a colored rectangular pattern.	glue (hot glue)	TPU	MSc
22	2021	Y.L	**Untitled**	Hats	A short, rounded hat with a wavy pattern in the skirt.	no assembly	TPU	MSc
23	2021	S.L	**Middle Eastern VEST**	Tops	This design is inspired by a unique second-hand vest the student owns, which was replicated with an oriental pattern.	welding - soldering iron	TPU	MSc
24	2021	G.L	**2D to 3D hat**	Hats	A hat made of rigid and flexible rings with a chevron pattern around the center. The stiffness	no assembly	TPU	MSc

					versus flexibility is achieved by adjusting the chevron angle.			
25	2021	I.M	**My wallet**	Bags	A transparent pocket-sized wallet made by folding a single piece of printed mesh. The mesh is inserted into the Nylon sheet protector and, with 4 folding, creates my 2 pockets, a standard card-size wallet.	no assembly	TPU	MSc
26	2021	N.R	**Digital lace gloves**	Accessories	Fingerless gloves, printed in two colors on tulle (white and transparent). The gloves incorporate a Fibonacci sunflower pattern.	sewing machine	TPU	MSc
27	2021	T.N	**Gcode Sandal**	Footwear	A sandal consists of a sole and a strap. The sole is constructed of two materials of different colors: the outer layer using PET and the insole using TPU. Both the sole and strap are based on a multidirectional grid.	glue (hot glue)	TPU	PhD
28	2021	I.P.N	**Fringe skirt**	Skirts	A skirt made with chevron-like panels stitched to the fabric. In the bottom panels, printed strands are loose, creating fringes.	hand sewing	TPU	MSc
29	2021	D.K	**3D Sleeve**	Accessories	A sleeve-like structure with a wavy silhouette and a twisted pattern is printed upright on the printer using under-extrusion to create perforations in the pattern.	no assembly	TPU	PhD
30	2021	N.R	**3D Glove**	Accessories	A mesh glove printed on the toe side, spread flat and apart. Additional pieces are connected between the fingers to produce volume of the glove.	glue	TPU	MSc
31	2021	R.S	**Pangolin Finger UV Protectors**	Accessories	A ring-like accessory designed to block UV light from the skin during the nail design process. The design allows for bending the finger with a mechanism inspired by a pangolin shell. The top part of the device is rigid, while the bottom is based on a diamond grid mesh.	glue	TPU	MSc
32	2021	M.S	**Untitled**	Face wear	A glasses frame with an exaggerated popish style design. The frame is based on a circular array of overlapping circles, and the temple is based on a linear array of the same circles.	mechanical connection	TPU	MSc
33	2021	S.A	**Untitled**	Tops	A vest made of separate panels that can be mechanically connected to form a garment. Some panels are composed of a grid of circles and others from a grid or square; the circles can be interlocked with the square to generate a mechanical connection.	mechanical connection	TPU	MSc
34	2022	A.A	**Lace Bracelet**	Accessories	A bracelet printed upright using under-extrusion to generate a perforation pattern. The base is integrated with extended arcs to create a decorative flower-like appearance.	no assembly	PLA	PhD
35	2022	A.B.Y	**PrintOnHole**	misc.	A method for fixing holes in garments by overlaying them using 3D printing. The process is based	Unspecified	TPU	MSc

					on a 3D scan to position and attach the print patches.			
36	2022	S.B	**LaceTop**	tops	A lace top bralette featuring a band and cup decorations with a pattern of overlapping circle and flower motifs. The straps are made with a diamond grid pattern that allows for even distribution of expansion force. Printing onto a mirror creates transitions between glossy and matte finishes.	Ironing with a backing paper	TPU	MSc
37	2022	L.G	**Replaceable strap slider shoes**	footwear	A slider shoe with replaceable straps. The design features side connectors that allow switching between different straps. The straps include various patterns: a grid, diamonds, and Voronoi. The sole is printed based on a grid pattern with varying density.	mechanical connections	TPU	Undergraduate
38	2022	A.G	**Untitled**	misc.	A Voronoi pattern printed onto racing shorts to provide stabilization, support, and localized compression.	no assembly	TPU	MSc
39	2022	N.D	**BridgeSlicer Hats production**	Hats	A series of perforated hats based on the bridgeSlicer tool.	no assembly	PLA	PhD
40	2022	Y.B	**Victoria's secret**	tops	A bra inspired by Roman patterns. Printed on paper. Printed at small scale	overlay printing - printing onto paper	PLA	MSc
41	2022	E.M	**Untitled**	footwear	A soccer sock sleeve with holes based on a magnetic grid pattern. The socks have large holes in them to imitate the holes soccer players make in their socks to relieve pressure.		TPU	MSc
42	2022	N.A	**Hat/kippa**	Hats	A hat that is based on printing with a height gap between the printed materials and the nozzle. The gap causes the material to droop and generate random complex morphologies.	no assembly	TPU	Undergraduate
43	2022	A.S	**The custom clutch**	bags	A clutch bag composed of 5 panels stitched together, the panels are made of overlapping zigzag panels with an organic pattern printed on top.	hand sewing	TPU	Undergraduate
44	2022	K.S	**Kraus eyewear**	face wear	A pair of glasses in a polygonal shape made with a twisted circular grid. The glasses are assembled using printed mechanical connectors.	mechanical connection	PLA	MSc
45	2022	N.S	**Printed skirt**		A full-size skirt made of panels with a triangular grid stitch together for a contoured look, using fabric that appears to have a glossy finish. A zipper is sewn at the back for ease of entry.	sewing	TPU	Undergraduate
46	2022	S.S	**Stretchable strap for a smart watch**	Accessories	A sporty watch strap inspired by an accordion with a pattern that enables flexibility.	no assembly	TPU	Undergraduate
47	2022	Y.S	**Earrings**	Accessories	A set of earrings and a pendant inspired by vintage jewelry. The design is based on a freehand sketch, which is offset multiple times and translated into a toolpath.	no assembly	PLA	MSc

48	2023	M.A. K	**Granny squares**	bags	a tote bag made of connected squares inspired by quilt blanket		PLA	MSc
49	2023	J.A.S	**NeedlePrints**	Accessories	A necklace and earrings with patterns and motifs inspired by traditional Arab embroidery designs.	Metal ring	PLA	MSc
50	2023	N.A	**Hello yellow**	tops	A vest top with interwoven sine pattern	Hand Sewing	TPU	Undergraduate
51	2023	O.A	**Kerchief**	Hats	a Kerchief basded on hexagonal grid with a shutter-like hole sizes transitions	Mechanical connections	TPU	PhD
52	2023	S.A	**Veiled fascinator hat**	Hats	a fascinator hat with a flexible faux-Sinamay brim	Ironing with a backing paper	TPU	MSc
53	2023	A.B	**Mandala top**	Tops	A ring shape shawl with a diamond grids pattern	Metal ring	TPU	MSc
54	2023	A.J	**3D Printed Ergonomic Wrist bracelet**	Accessories	3D Printed Ergonomic Wrist Bracelet - based on 3D printed spacer fabrics	No assembly - single part + strap	TPU	MSc
55	2023	L.G	**3D printed net-lace top with printed joints**	Tops	A lace top with flower motifs embedded in the printed panels.	Mechanical connections	TPU	MSc
56	2023	K.G	**Grasshopper Codded Collar**	Accessories	a collar with a wavy pattern	No assembly - single part	TPU	MSc
57	2023	I.D	**Calligraphy kerchief**	Hats	Fabric shawl with attached 3D-printed panels featuring geometric patterns embedded with Arabic calligraphy letters.	Hand Sewing	TPU	MSc
58	2023	L.T	**Untitled**	tops	a vest with chainmail-like pattern	3D pen	TPU	MSc
59	2023	A.C	**Robotic printing wood en textile**	Misc.			Wood paste	PhD
60	2023	M.L	**Décolletage net**	Accessories	A decorative net-like piece to connect a collar with a dress.		TPU	MSc
61	2023	U.L	**Untitled**	Accessories	A necklace and earrings with circular plates filled with sine wave patterns.	Metal ring	PLA	Undergraduate
62	2023	M.M	**Untitled**	Accessories	A decorative flower-like structure that can be embedded in a necklace or a ring.	Thermal bonding	PLA	Undergraduate
63	2023	A.M	**ArabesCap**	Hats	Islamic prayer hat - with Islamic - like patterns	No assembly - single part	PLA	Undergraduate
64	2023	R.M	**Untitled**	Accessories	A visor that can also be worn as a necklace, designed with a deformed grid of dancer motifs.	3D pen	TPU	MSc
65	2023	S.N	**Untitled**	Tops	A printed corset/belt based on a hexagonal grid	Hand Sewing	TPU	Undergraduate
66	2023	S.S	**Podpockets**	Accessories	a bracelet with a pocket to store the air pods case during workout.	hand sewing	TPU	Undergraduate
67	2023	A.S	**i3D**	AccAirPries	An iWatch band composed of wavy lines	no assembly - single part	PLA	Undergraduate
68	2023	S.S	**3D Printed geographic bucket Hat**	Hats	a bucket Hat made with a grid net and a topographic map-like pattern.	Welding - soldering iron	PLA	MSc
69	2024	S.A	**Purim hat**	Hats	a funny looking small clowns' hat	no assembly	PLA	MSc

70	2024	A.Z	**3D Printed tote bag**	bags	A tote bag composed of interlocking lave panels. The design is based on a ring connecting four flower-like panels that can be further connected to form a continuous fabric.	Mechanical connections	TPU	Undergraduate
71	2024	O.A	**Tembel**	Hats	A short brim hat made out of 6 panels with embedded flaps. The panels are stitched together using the flat to create the hat structure.	Mechanical connections	TPU	PhD
72	2024	Z.B	**Mesh gloves**	Accessories	Printing a custom-fit glove made of two layers welded together in a single run. First, the bottom layer is printed, then the printing pauses and a separator is placed. Next, the top layer is printed, and the separator is removed.	assembly using the printer	TPU	MSc
73	2024	R.B	**Parametric shoe**	footwear	a show with a printed sole in organic pattern	glue	PLA	MSc
74	2024	D.J	**Abayee's harness**	Pets	A dog harness composed of small square panels connected with H-shaped connectors.	mechanical connections	PLA	Undergraduate
75	2024	E.H	**Cap hat**	Hats	A cap hat made of connected panels featuring a pattern of a woman's figure.	printing onto fabric	PLA	MSc
76	2024	E.H	**Sling bag**	bags	a sling bag folded - ironed onto fabrics	Ironing with a backing paper	PLA	MSc
77	2024	A.W	**Parametric Jewelry - échec collection**	Accessories	a bracelet and ring designed with a swiveling pattern.	No assembly - single part	PLA	MSc
78	2024	E.K	**Digitally crafted niqab**	Accessories	In this project, a niqab for young girls is printed. It features delicate floral and decorative motifs. The base fabric is inspired by traditional tatting lace techniques. The fabric is created through a customizable grid system that mimics the delicate interlacing of threads. The open nature of the grid design allows connector buttons to be inserted into its holes. This enables the attachment of decorative flowers, offering each wearer a customizable and personal touch.	mechanical connections	TPU	Undergraduate
79	2024	Y.B	**No Shade**	Hats	A visor hat with a curvy silhouette.	glue	PLA	MSc
80	2024	S.Y	**Patagonia pouch**	bags	a pouch with Patagonia graphics. 3D printing onto textile pattern pieces	Sewing machine	TPU	MSc
81	2024	Y.L.	**Sead3**	bags	a small beach bag with a Voronoi pattern	glue	TPU	MSc
82	2024	A.S	**Auxetic knee protector**	misc.	A knee protector with an auxetic structure that can open and close to accommodate the knee curvature.		TPU	MSc
83	2024	O.O.	**Apro-Moiré**	tops	an apron with a grid pattern and an image of hamster created using Moiré pattern	Welding - soldring iron	TPU	Undergraduate
84	2024	T.P.G.	**3D printed fabric wallet**	bags	A credit card-sized wallet made from a folded flat panel with a deformed grid pattern. Two Metal connectors are attached for locking.	no assembly - folding	TPU	MSc

85	2024	A.R	**The TPShoe 95**	footwear	A single-piece 3D printed shoe upper with integrated flaps that fold into a 3D shoe shape.	hand sewing	TPU	Undergraduate
86	2024	D.S.	**Wide brim Hat**	Hats	A long-brimmed hat with a straw hat – inspired pattern.	glue	PLA	Undergraduate
87	2024	S.L.	**VeilX**	Hats	A protective net attached to a hat, inspired by beekeeping veils and outdoor gear. The net is connected to a hat using metal rings.	Metal ring	PLA	MSc
88	2025	O.A	**Waist belt**	Accessories	A belt or corset featuring a 3D printed pattern that mimic knitting structures. The belt closes by interlocking an array of printed loops from both sides into each other	Ironing with a backing paper	TPU	MSc
89	2025	S.O	**Kirigami bag**	bags	A rounded-shape printed mesh with incorporated slits to enable Kirigami folding once lifting the bag from the handles.	No assembly - single part	TPU	MSc
90	2025	S.A	**Vitamin - G**	bags	A bag made with mesh-like panels featuring an embedded pattern of oranges, connected using a 3D pen. Trims and added using a sewing machine,	3D pen	TPU	MSc
91	2025	Y.B	**3D Pow**	bags	A wallet created from a printed sheet with embedded fold lines and a graphic pattern. In a specific sequence, the sheet is folded into a wallet shape	No assembly - single part	TPU	MSc
92	2025	Y.H	**Mandala collar**	Accessories	A circular collar based on a lotus leaf pattern. The collar is printed flat with the petals lifted and supported in mid-air. The structure is flexible, allowing the collar to stretch over the head and curve outward when worn.	No assembly - single part	TPU	MSc
93	2025	H.Y.	**3 corners hat**	Hats	A hat featuring a Star of David pattern.	Glue	PLA	MSc
94	2025	L.Y.	**Untitled**	Pets	A dog collar made using 3D printing onto knitted fabrics. Two patterns are printed: one pattern is based on overlapping circles, and the other pattern is composed of different-sized squares.	Printing on knitted fabric	TPU	MSc
95	2025	N.Y	**Hand fan**	Accessories	A hand fan with a printed textile pattern of arcs. The printed textile is locked by 3D-printed spines, which enable the fan to open and close. The textile pattern was also used to create earrings.	Glue	TPU, PLA	MSc
96	2025	S.C	**G-code to wear- bag**	bags	A bag composed of star-shaped pieces connected with printed circle connectors	mechanical connections	TPU	MSc
97	2025	S.L.	**Plaid market**	bags	A market bag with a plaid-like pattern made with panels of two colors of filament.	Sewing machine	TPU	MSc
98	2025	S.O.	**Mandala wear**	Pets	A dog harness composed of 4 lacey bands, connected using metal rings and latches.	Metal ring	TPU	MSc
99	2025	E.P.	**Grass head**	Hats	A hat or wig incorporates a printed textile mesh and soil. Sprouts are growing in the soil and emerge through the mesh, creating a humorous appearance of a human grass head.	hand sewing	TPU	MSc

100	2025	E.K	**Stitch the print**	Accessories	A 3D printed technique for assembling large-scale soft textiles by using the 3D printer as a sewing machine. The connected pieces form a quilt-like shawl.	using the printer to assemble	TPU	MSc
101	2025	M.K	**Computational bralette**	tops	A fully 3D printed lace bracelet with different density zones to match the shaped area of the body and strength constraints.	using the printer to assemble	TPU	MSc